PhysAgent: Physics-Informed SDevice Code Generation via Ontology-Guided LLM

Shinbin Kim*
Dept. of Semicond. Eng.
POSTECH
Pohang, Rep. of Korea
sinbin0701@postech.ac.kr

Giung Kim*
Dept. of Semicond. Eng.
POSTECH
Pohang, Rep. of Korea
giung0408@postech.ac.kr

Jiwon Lee†
Dept. of Semicond. Eng.
POSTECH
Pohang, Rep. of Korea
jiwonlee@postech.ac.kr

Dae-Hwan Kang†
Dept. of Semicond. Eng.
POSTECH
Pohang, Rep. of Korea
daehwankang@postech.ac.kr

***Abstract*—As semiconductor scaling reaches its limits, Technology Computer-Aided Design (TCAD) simulation becomes essential for Design-Technology Co-Optimization (DTCO). However, the core challenge lies in configuring SDevice—the physics-and-solver specification stage of Sentaurus TCAD—rather than in device structure generation. We propose PhysAgent, which builds a Sentaurus TCAD guide-based ontology and integrates literature-driven domain knowledge to generate physically grounded SDevice files without relying on synthetic training data.**

***Keywords*—TCAD, ontology, physics-informed modeling, parameter file generation, domain knowledge, semiconductor materials**

I. Introduction

Recent studies have applied LLMs to automate TCAD code generation, primarily targeting structure files written for SDE (Sentaurus Structure Editor). These approaches rely on supervised fine-tuning (SFT) with synthetic datasets constructed by varying numerical parameters in a small number of baseline scripts [3]–[6]. However, this limits generalization to unseen device architectures or materials. More critically, none adequately addresses physics model selection or parameter file generation for SDevice—the core of physically meaningful TCAD simulation. Even AgenticTCAD [6], which generates SDevice code, uses only 2–3 fixed physics configurations and omits parameter files entirely.

We propose PhysAgent, which constructs a Sentaurus TCAD guide-based ontology encoding physical models, equations, and their interdependencies, complemented by literature-driven material-specific knowledge. By providing this structured context to general-purpose LLMs, PhysAgent generates physically grounded SDevice files with explicit reasoning about model selection and parameter settings—without domain-specific fine-tuning.

II. Related Work

TCAD simulation consists of SDE for device geometry and SDevice for physics models and solver settings. While LLMs have been applied to SDE code generation through fine-tuning or in-context learning, SDevice configuration—selecting and parameterizing models such as mobility, recombination, and bandgap narrowing—remains unaddressed due to its high expertise requirements.

*These authors contributed equally to this work.
†Corresponding authors.

III. Proposed Framework

PhysAgent combines a TCAD physics ontology and literature-driven domain knowledge to guide LLM-based SDevice file generation. The constructed ontology is publicly available as an interactive web platform [7].

A. TCAD Physics Ontology

We construct a directed graph from the Sentaurus TCAD guide with three node types (*equation*, *parameter*, *constant*) and three edge types (*input*, *specialCase*, *pairwise*), capturing inter-model dependencies such as bandgap narrowing affecting intrinsic carrier density and recombination rates (Table I).

B. Ontology-Guided Generation

Given a target device and material, PhysAgent accesses the ontology through an MCP (Model Context Protocol) interface that exposes three classes of tools: one returning a node/edge map of the graph, one returning SDevice keywords and syntax for a selected node, and one returning material-specific parameter values together with their source citations drawn from a curated set of TCAD publications (e.g., Masetti, Klaassen, Roschke, Pernot). Since the full ontology spans hundreds of thousands of tokens, direct prompt injection is impractical; the LLM instead follows a selective retrieval loop—reading the overview, fanning out parallel syntax queries for the relevant models, pulling references for parameter values, and composing the Physics block and `.par` file. This shrinks per-query context by roughly 6× relative to a full-ontology dump while preserving model coverage, and each parameter in the generated `.par` inherits the citation returned by the reference tool, enabling parameter-level provenance (Fig. 1).

C. Ontology Expansion and Validation

Unlike SFT-based approaches requiring retraining for new models or materials, PhysAgent uses a directional synchronization workflow with three artifact layers—Spec, Plan, and Impl—where changes propagate automatically from the source of truth while reverse modifications require explicit approval, ensuring incremental expansion without retraining.

IV. Experimental Validations of PhysAgent

PhysAgent targets the physics specification stage—the SDevice `Physics` block and `.par` file—while structure (`.tdr`), solver, and bias schedule remain fixed, orthogonal to prior structure-generation work [3], [5]. Three generators share the same base LLM and prompt: *Vanilla* (spec only), *LLM+Guide* (spec + Sentaurus manual), and *PhysAgent* (spec + ontology + literature). We validate three claims.

Successful TCAD Execution. On a Si nMOSFET we compare three modes (Vanilla, LLM+Guide, PhysAgent) across three prompt levels (L1–L3: explicit model list, categories, intent only), three trials each. Vanilla and LLM+Guide produce characteristic syntactic errors—invented model names, undefined parameters, malformed material blocks—lacking any structural anchor, whereas PhysAgent copies directly from typed ontology nodes at 48× smaller prefill than LLM+Guide. Across L1→L3 model coverage tracks prompt specificity: which physical mechanisms the deck selects depends on what the prompt asserts, making phenomenon-level prompting central to the simulation's analytical scope (Fig. 2).

Literature-Grounded Parameter File Generation.

Phenomenon-Driven Model Selection and Deck Construction.

V. Conclusion

We presented PhysAgent, which generates physically grounded SDevice files by combining a TCAD guide-based ontology with literature-driven domain knowledge, enabling general-purpose LLMs to support expert-level TCAD modeling without synthetic training data.

TABLE I
TCAD PHYSICS ONTOLOGY COMPONENTS.

Component	Type	Description
Node	`equation`	Physical model equations
	`parameter`	Constants, variables, control flags
Edge	`uses`	Equation consumes a parameter
	`controls`	Flag determines model activation
	`routed_by`	Conditional branching path
Group	`fundamental`	Core physics (bandgap, n_i)
	`physicsModel`	Specific models (Masetti, SRH)
	`auxiliary`	Corrections and helpers
Relation	`exclusive`	Choose one among alternatives
	`compatible`	Can coexist
	`incompatible`	Cannot be combined

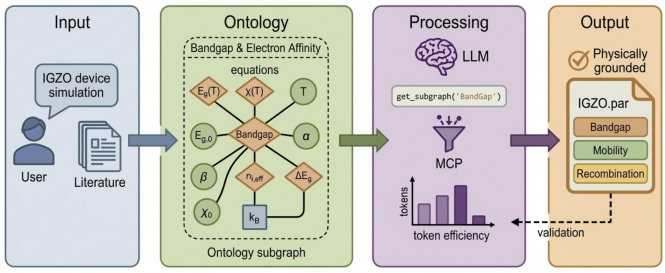

Fig. 1. PhysAgent framework: LLM agents access the ontology graph via MCP tool calls to generate physically grounded parameter files.

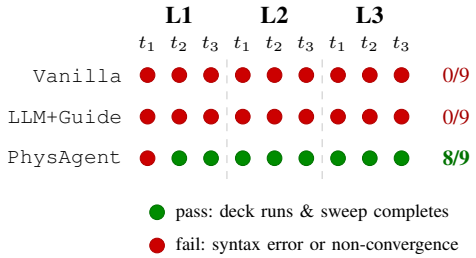

Fig. 2. Per-trial TCAD execution on Si nMOSFET across prompt-detail levels L1/L2/L3 and trials t_1–t_3. A trial passes only if Sentaurus parses the generated deck without syntax errors and the Id–Vg sweep completes. Vanilla and LLM+Guide fail on every trial—predominantly at parse time—while PhysAgent passes 8/9.

[Placeholder: Model selection accuracy and parameter match rate of PhysAgent vs Sentaurus library across 34 semiconductor materials]

Fig. 3. Reproduction of the Sentaurus library: PhysAgent vs library across 34 materials.

[Placeholder: Id–Vg overlay of PhysAgent-generated a-IGZO TFT vs digitized published measurement]

Fig. 4. Extension beyond the library: a-IGZO TFT Id–Vg from PhysAgent-generated deck vs published measurement.

References

[1] Z. Ji *et al.*, "Toward reliability- and variability-aware design-technology co-optimization at advanced nodes," *IEEE Trans. Electron Devices*, vol. 71, no. 1, pp. 138–150, 2024.

[2] Synopsys, "Sentaurus TCAD," 2025. [Online]. Available: https://www.synopsys.com/manufacturing/tcad.html

[3] L. M. L. Nguyen, A. Lu, and H. Y. Wong, "TCAD structure input file generation using large language model," in *Proc. SISPAD*, 2024, pp. 1–4.

[4] Z. Zhang, S. Nassif, and M. Tahoori, "ChatTCAD: Leveraging large language models for automated TCAD simulation file generation," in *Proc. ISVLSI*, 2025, pp. 1–6.

[5] J. Dou *et al.*, "MALTS: A multi-agent large language model-based layout-to-TCAD structure synthesizer," in *Proc. ISEDA*, 2025, pp. 795–800.

[6] G. Fan *et al.*, "AgenticTCAD: A LLM-based multi-agent framework for automated TCAD code generation and device optimization," *arXiv:2512.23742*, 2026.

[7] PhysAgent Team, "TCAD Physics Ontology," 2026. [Online]. Available: https://ontology-tcad.vercel.app/